Course Code : ECT 252 **GVHW/RW – 23 / 1428**

Third Semester B. Tech. (Electronics and Communication Engineering) Examination

DIGITAL SYSTEM DESIGN

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions carry marks as indicated.

(2) Assume suitable data wherever necessary.

1. Simplify the following Boolean equation using k-map and then implement the simplified equation using NOR gates.

$$Y = \Sigma m(0, 1, 3, 4, 6) \cdot d(2, 10, 15)$$ 10(CO1)

2. (a) Design a 16-bit tri-state buffer using dataflow modeling. 5(CO2)

   (b) Explain the net types in Verilog with the help of an example. 5(CO2)

3. (a) Write a UDP using Verilog for a gate that can detect presence of two or more than two 1's at its 5 inputs. 6(CO4)

   (b) Write a Verilog module that instances the UDP, designed in question 3 (a). 4(CO4)

4. (a) Design a D flip-flop using Verilog. 4(CO4)

   (b) Design a SISO shift register using D flip-flop, designed in question 4 (a). 6(CO4)

5. Explain architecture of PAL and implementation of Boolean equation using it. 10(CO5)

6. (a) Enlist typical of characteristics of CMOS logic family. 5(CO3)

(b) A list of performance parameters of logic family is provided as follows. Indicate the best performing logic family for every parameter along with typical value of parameter :

(i) Propagation delay,

(ii) Fan-out,

(iii) Fan-in,

(iv) Power dissipation,

(v) Noise margin. 5(CO3)